

Airway Resistance

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Learning Objectives (2)

After completing this brick, you will be able to:

- 1 Explain how airway resistance is affected by airway radius, turbulent flow, lung compliance, and lung volume (including La Place)
- 2 Explain the pressure-volume (compliance) curves for the lungs, chest wall, and respiratory system (including the FRC)



CASE CONNECTION

JW, a 66-year-old male and former smoker, presents to his primary care physician with worsening shortness of breath and a chronic productive cough. He reports shortness of breath, especially during exertion, and notes that climbing stairs has become increasingly challenging. On physical exam, he has a prolonged expiratory phase and scattered expiratory wheezes. Pulse oximetry

shows an oxygen saturation of 91% on room air. A chest radiograph shows flattened diaphragms and hyperinflated lungs. Pulmonary function tests reveal an increased total lung capacity (TLC), reduced forced expiratory volume in 1 second (FEV₁), reduced FEV₁/FVC ratio, and elevated residual volume (RV). He is diagnosed with chronic obstructive pulmonary disease (COPD), likely emphysema-predominant.

What mechanisms lead to increased airway resistance in JW's lungs? How do changes in compliance and dynamic airway behavior affect his breathing pattern and effort?

[GO TO CONCLUSION](#) ↓

Introduction: What Determines Airway Resistance (LO1)?

Breathing involves moving air into and out of the lungs through a system of branching airways. **Airflow** is determined by the pressure difference driving flow relative to the resistance within the airways, as described by the equation: **Flow (Q) = $\Delta P / R$** . **Airway resistance** is the opposition to airflow.

Airway Radius: Small Changes, Big Effects

The most powerful determinant of airway resistance is the radius of the airway. It can change from moment to moment, depending on lung volume, airway smooth muscle tone, and pathology. Poiseuille's law governs the relationship between airway radius and resistance, stating that resistance is inversely proportional to the fourth power of the radius ($R \propto 1/r^4$). This relationship has profound clinical implications: a 50% decrease in airway

radius results in a 16-fold increase in resistance. Even small changes in airway caliber can dramatically alter breathing mechanics.

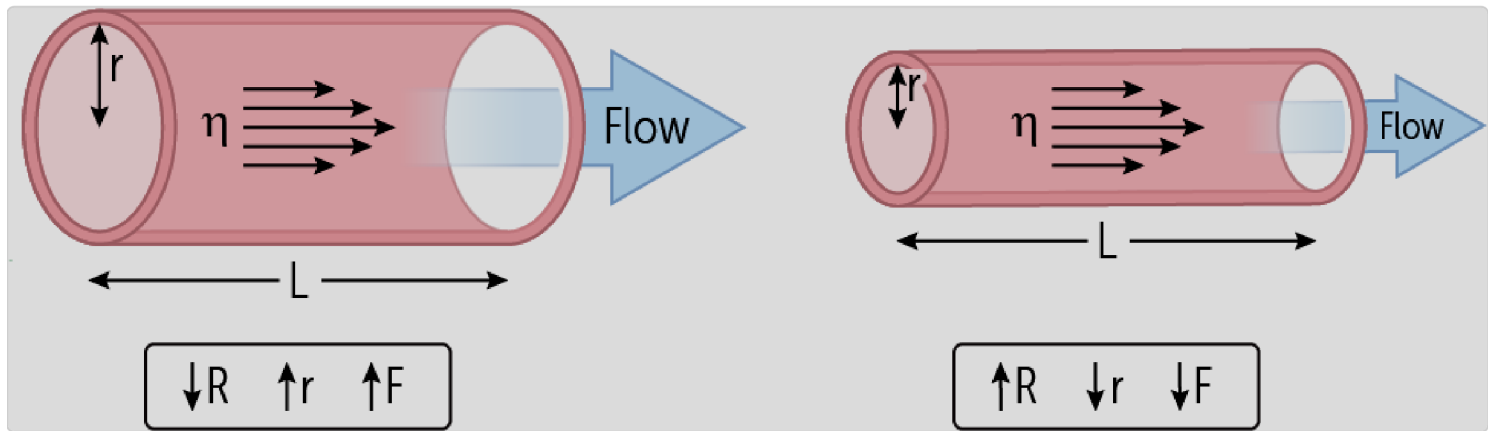


Figure 1 Effect of airway radius (r) on resistance (R) and laminar airflow



CLINICAL CORRELATION

In conditions with airway narrowing (decreased airway radius) and high resistance, more effort is required to breathe. Examples include asthma (inflammation and increased mucus production), bronchospasm, or chronic obstructive pulmonary disease (COPD), which can make breathing difficult, particularly on exhaling. This reduced ventilation limits the exchange of oxygen and carbon dioxide in the alveoli.

Therapeutic interventions such as beta-agonist bronchodilators and anti-inflammatory medications work by increasing airway radius through smooth muscle relaxation and reduction of mucosal edema, respectively. The effectiveness of these treatments directly reflects the powerful influence of radius on airway resistance.

Regulation of Airway Radius occurs by two main factors:

- **Autonomic nervous system.** Contraction of the bronchial smooth muscles reduces the airway radius because these muscle cells surround the airway. Contraction is caused mostly by parasympathetic nervous system innervation, mediated by cholinergic fibers. Conversely, sympathetic stimulation of adrenergic β_2 receptors results in bronchial smooth muscle relaxation, increased airway diameter, reduced resistance, and increased airflow.
- **Lung volume.** The total lung volume also influences resistance; as the lungs inflate, the airways are pulled open by radial traction of the lung tissues, reducing airway resistance. Therefore, during inspiration and as lung volumes increase, airway resistance decreases and airflow increases (more later in this brick).

ACTIVE LEARNING QUESTION

A 14-year-old child experiencing an acute asthma attack is treated with albuterol, a sympathetic β_2 -agonist that relaxes the bronchial smooth muscles. This treatment results in a two-fold increase in the diameter of the child's airways. Which of the following best describes the resistance in the patient's airway after treatment compared with that before receiving albuterol?

- A. Resistance decreases by a factor of 2
- B. Resistance decreases by a factor of 4
- C. Resistance decreases by a factor of 16
- D. Resistance decreases by a factor of 32

Explanation (requires correct answer)

Cross-sectional area of the bronchial tree

In addition to radius at a single point in the airways, we also consider the **total cross-sectional area** of the airways at different levels or generations of the bronchial tree (Figure 2). While individual bronchioles are small, with small radii, the large number of them in parallel means their **combined cross-sectional area is quite large**, resulting in **low total resistance**. This is why the **small airways**, though narrower, contribute less to total airway resistance under normal conditions than the medium-sized bronchi. An analogy is the work to breathe through a large straw (upper airways), one straw (a single small airway), or several straws (large cross-sectional area and air flows in a parallel fashion).

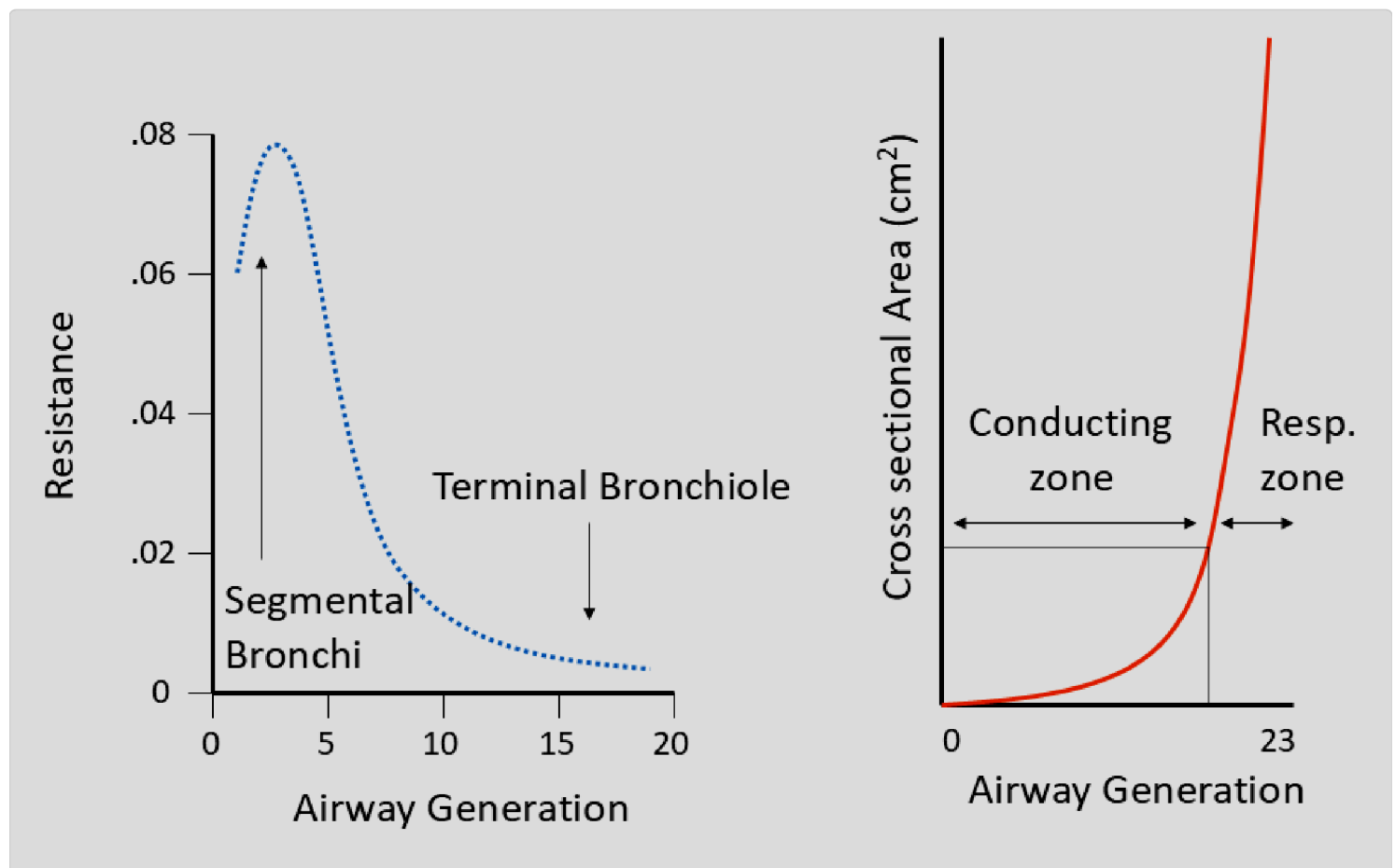


Figure 2. Airway resistance and cross sectional area of the airway generation

Flow Pattern: Turbulent vs Laminar

Airflow patterns significantly influence resistance: laminar flow is characterized by smooth, layered movement of air molecules, while **turbulent flow** features chaotic vortices and eddies ([Figure 3](#)). Laminar flow predominates during quiet breathing and in smaller airways. Turbulent flow occurs in larger airways, where the cross-section is large enough for disordered flow to occur. This is more noted during forced breathing or in the presence of an upper airway obstruction (e.g., a marble or with croup), the latter causing a sound called stridor ([sound link](#)).

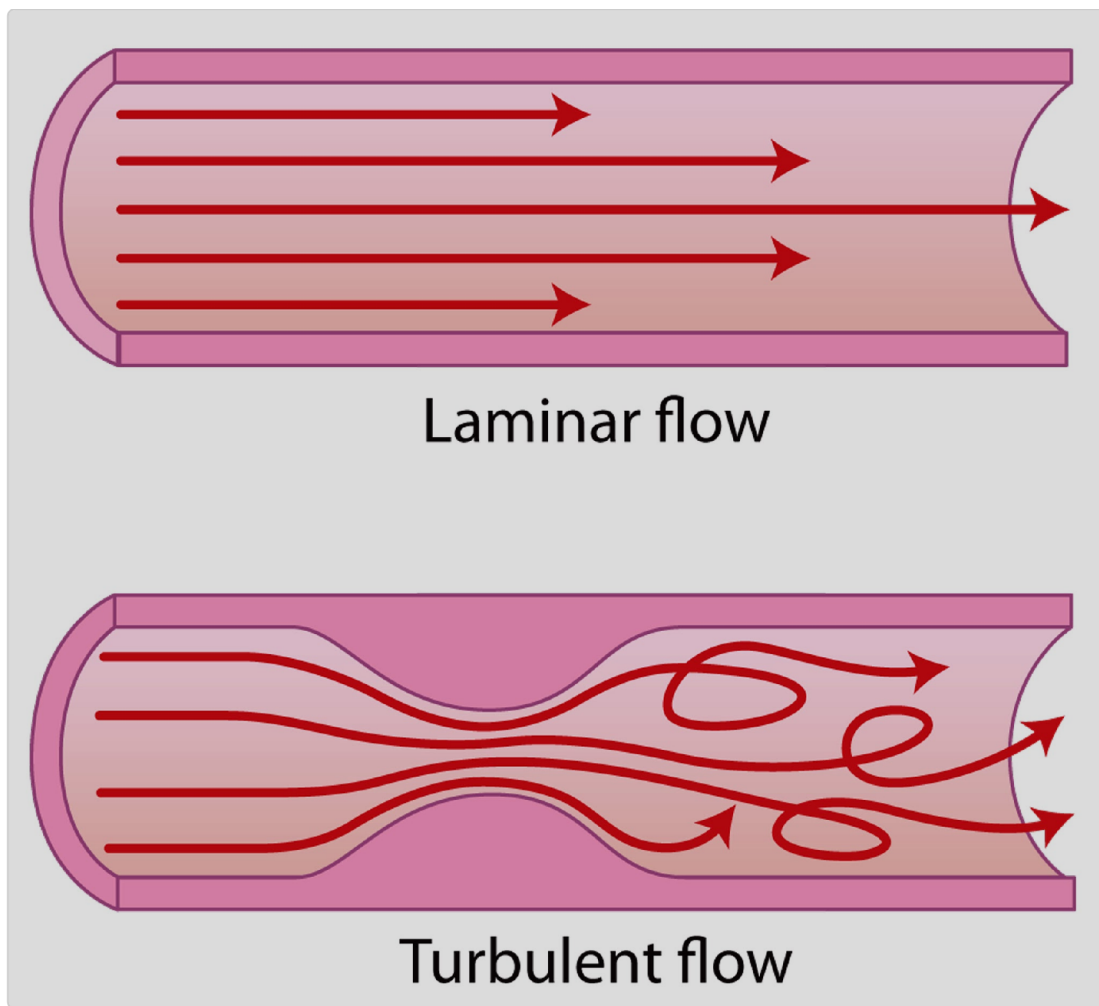


Figure 3 Laminar vs turbulent airflow

Turbulent flow substantially increases resistance compared to laminar flow, requiring higher pressure gradients to maintain the same airflow rate and increasing the **work of breathing** (muscle effort). The Reynolds number helps predict when flow will become turbulent; values above 2000 result in turbulent flow (equation not needed). Turbulence or airflow obstruction also causes chest sounds heard during auscultation. **Wheezes** are high-pitched, musical sounds produced by turbulent airflow through **narrowed small airways**, as in asthma ([sound link](#)) or COPD, where airway walls flutter during expiration. Rhonchi are low-pitched, snoring-like sounds ([sound link](#)) due to **airflow through mucus-filled bronchi**.

Lung Volume and Radial Traction

Radial traction is the mechanical force exerted by surrounding lung tissue on the airways and pulls them open during lung inflation ([Figure 4](#)). As lung volume increases, the expanding alveoli and surrounding connective tissue pull outward on the airway walls, increasing tension, widening the airways, and reducing resistance. This mechanism explains why airway resistance decreases with deeper inspiration and increases during expiration as lung volume falls.

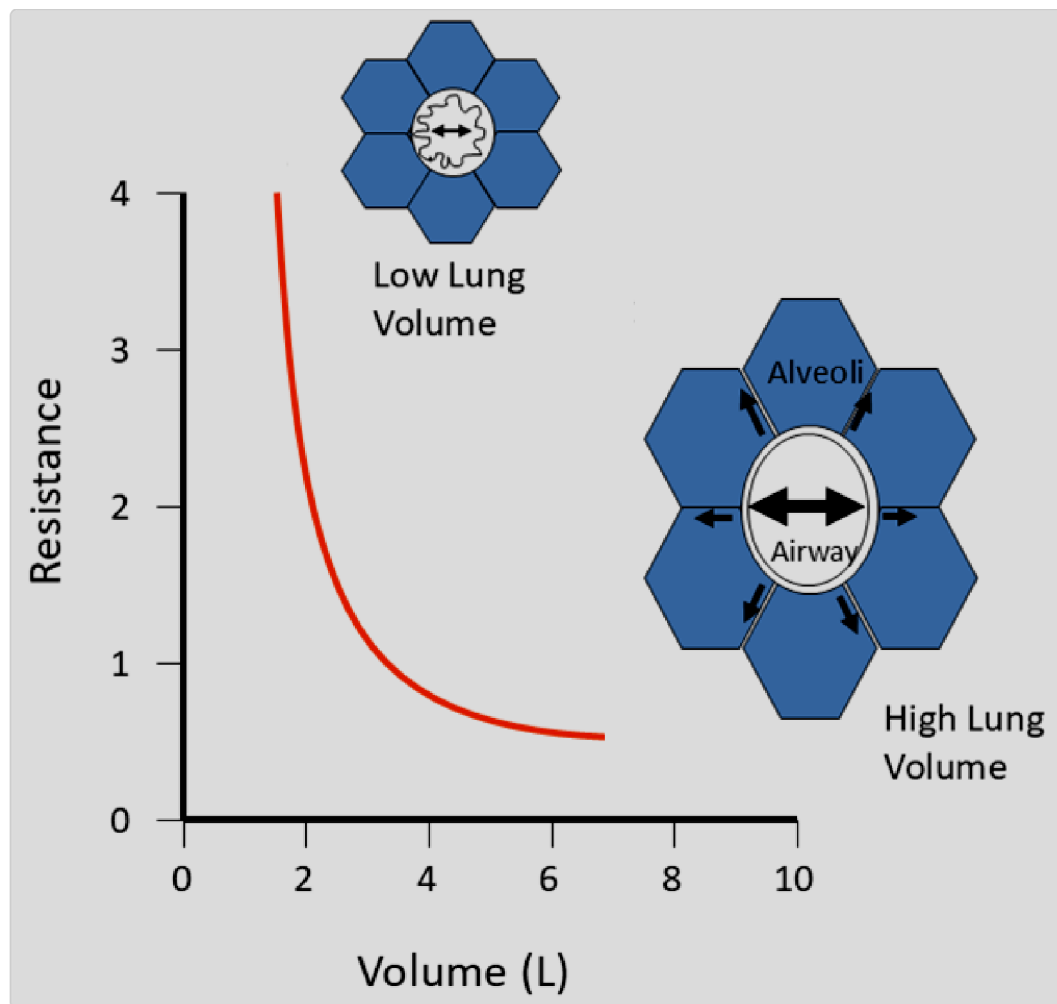


Figure 4. Effects of lung volume on radial traction and airway resistance

CLINICAL CORRELATION

The clinical importance of **radial traction** becomes apparent in diseases affecting lung elasticity.

In **emphysema**, destruction of elastic fibers weakens radial traction, allowing airways to collapse more readily during expiration, which contributes to air trapping in the lungs.

Conversely, in **pulmonary fibrosis**, while the stiff lung tissue may provide some structural support, the reduced lung volume/lung expansion limits the effects of

radial traction on airway patency (opening), and thus airway resistance is increased throughout the breathing cycle.

radius (r).

by 20%, as resistance is inversely proportional to the fourth power of Airway resistance increases by 16 times (2^4) when radius decreases

Dynamic Airway Collapse

Dynamic airway collapse occurs when intrathoracic pressure exceeds airway pressure during expiration, and presses the airways closed. The collapse typically affects small bronchi and bronchioles that lack cartilaginous support, creating a functional obstruction to airflow. In normal lungs, this occurs only at high expiratory flows because, during quiet breathing, airway pressure (blue color) remains above pleural pressure (cream color), and radial traction from alveoli keeps the airways open (Figure 5A, top). However, during forced expiration, intrapleural pressure rises sharply, increasing alveolar pressure and driving air down the pressure gradient from the alveoli

to the mouth (Figure 5A, middle). Since airway pressure falls toward zero at the mouth, there is a point in the airway when pleural pressure exceeds airway pressure, producing an **equal pressure point** where the airway narrows (“choke point”).

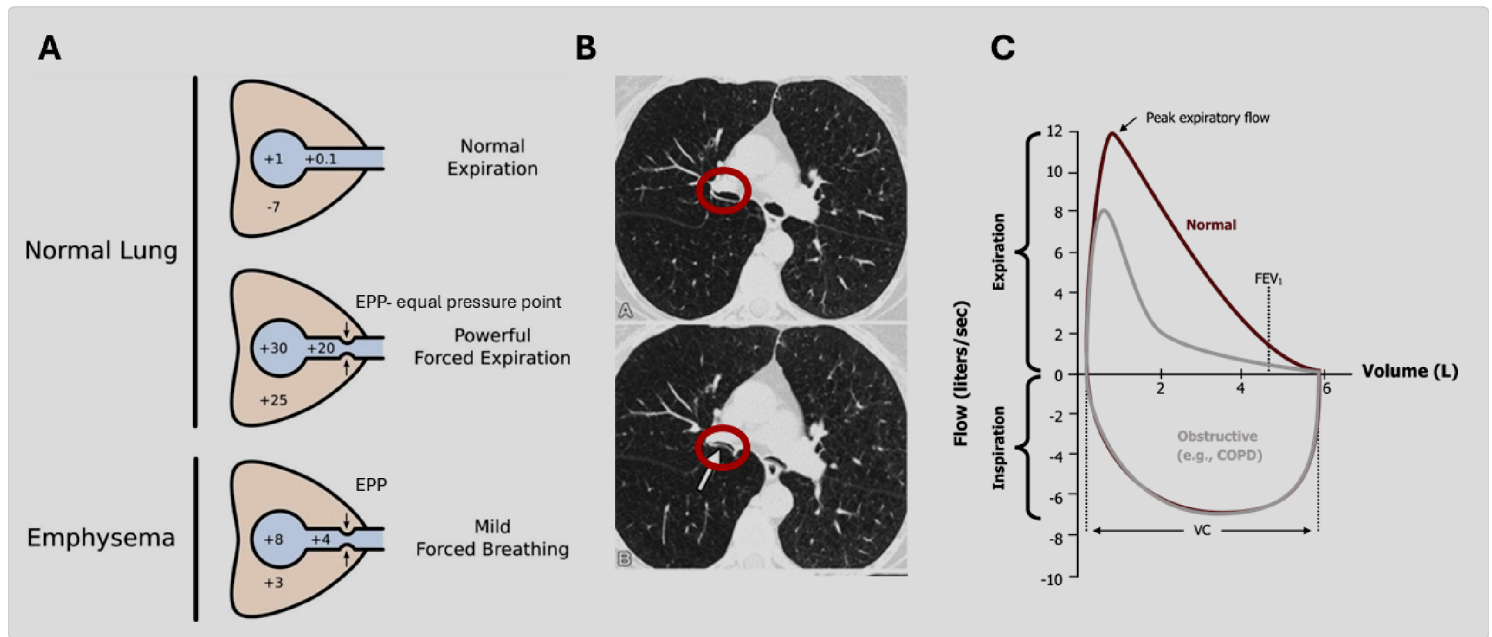


Figure 5. Dynamic airway collapse- illustration and in a CT and PFT

In emphysema, the loss of elastic recoil due to tissue destruction reduces the alveolar pressure generated and radial traction, so even mild expiration and lower intrapleural pressures result in airway collapse (Figure 5A, bottom). The CT image shows alveolar destruction and loss of support, and collapse of the bronchi between inspiration (top) and expiration (bottom, arrow, and circles (Figure 5B). The pulmonary function test (PFT) flow-volume loop shows the “scooped-out” expiratory pattern of obstructive disease, in which dynamic airway collapse and loss of radial traction reduce expiratory flow (Figure 5C).

ACTIVE LEARNING QUESTION

A 70-year-old male with emphysema presents to the emergency department with breathing difficulty. Of the following sets of parameters, which would be characteristic of this patient's respiration?

- A. $\uparrow$ compliance, $\uparrow$ resistance, $\uparrow$ turbulent flow
- B. $\uparrow$ compliance, $\uparrow$ resistance, $\downarrow$ turbulent flow
- C. $\uparrow$ compliance, $\downarrow$ resistance, $\downarrow$ turbulent flow
- D. $\downarrow$ compliance, $\downarrow$ resistance, $\downarrow$ turbulent flow

Explanation (requires correct answer)

How Do Compliance Curves for the Lungs, Chest Wall, and Respiratory System Relate to the FRC (LO2)?

Lung compliance (pressure volume curve)

Lung compliance describes the **ease with which the lungs expand** (distensibility) in response to a given pressure and is defined as the change in volume per unit change in pressure ($C = \Delta V / \Delta P$). A helpful analogy is comparing a water bottle to a plastic grocery bag: plastic grocery bag = high compliance because it inflates with little effort and pressure, water bottle = low compliance because it takes more effort to stretch it. A steeper slope indicates higher compliance (large volume change for small pressure change, or "compliance to stretch"). In comparison, a flatter slope indicates lower compliance (less volume change per pressure change, or stiffer lungs) **Figure 6**.

At low lung volumes compliance is low (stiffer lungs). Many alveoli are collapsed or near collapse, and require greater effort to inflate, and thus a large pressure change yields little volume change **Figure 6**. **At very high volumes**, near **total lung capacity** (TLC), the lung tissue resists further expansion and the curve flattens again, reflecting decreased compliance **Figure 6**. This is because the alveoli and elastic fibers are

maximally stretched. The **mid-volume** range, corresponding to normal tidal breathing, represents optimal compliance where the lungs expand most easily **Figure 6**. The degree and ease with which the lungs expand are the main determinants of the work of breathing.

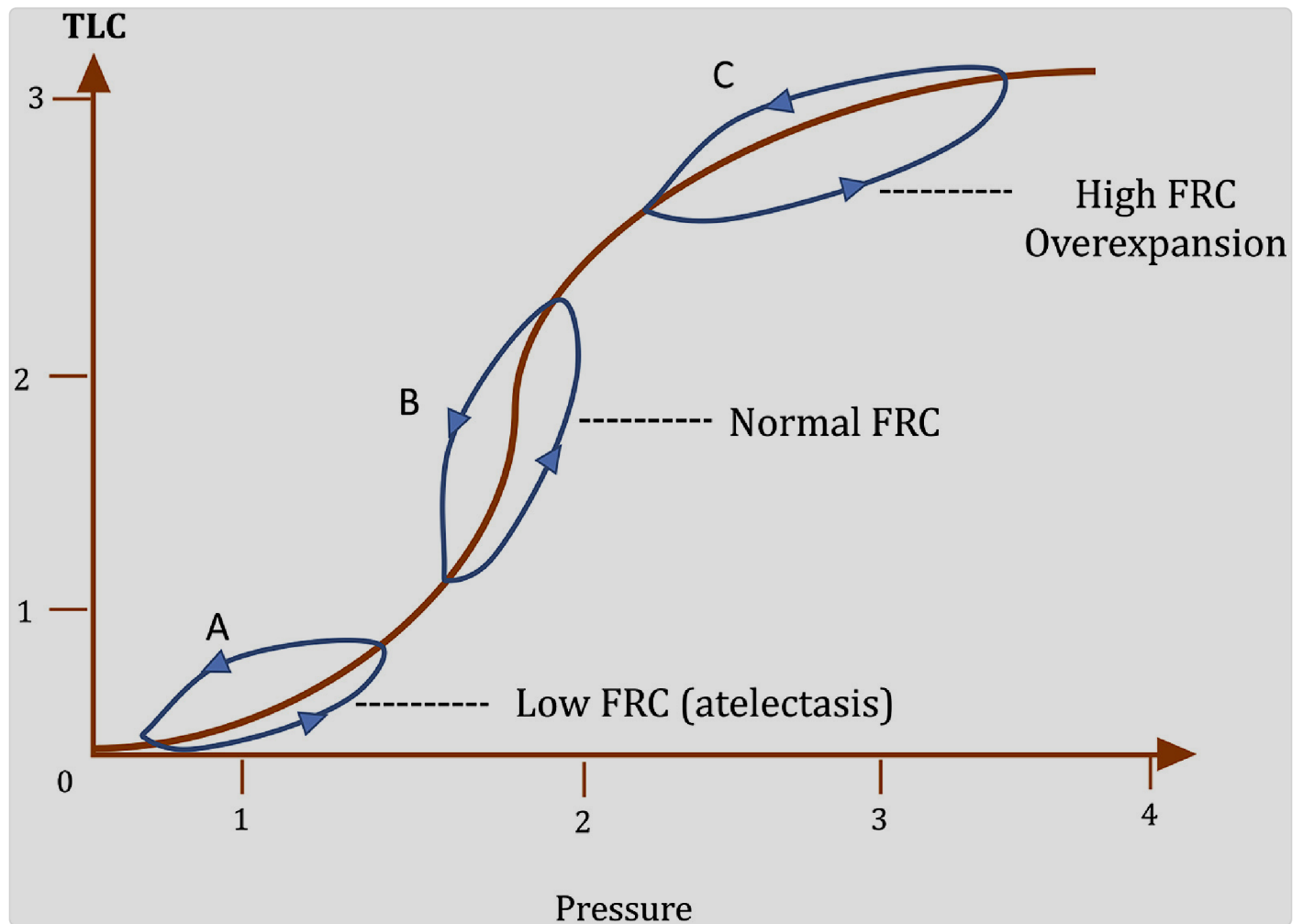


Figure 6. Lung compliance at different volumes

CLINICAL CORRELATION

Disease states cause predictable shifts in these curves that correlate with clinical presentations.

In **restrictive diseases** like pulmonary fibrosis or neonatal respiratory distress syndrome (RDS), the curve shifts downward and to the right → lower compliance. Thus a large change in pressure is needed for small change in volume. This makes it harder for patients to breathe through more-rigid lungs.

In **obstructive diseases** like emphysema and asthma, the curve shifts upward and to the left, → increased compliance due to loss of elastic tissue. For a small change in pressure, there is a large change in volume. Note that in these patients, reduced recoil and air trapping predispose to **dynamic airway collapse**.

distensible, elastic lungs.

more volume change per unit pressure change. This represents more
A steeper compliance curve indicates greater compliance, meaning

Determinants of lung compliance:

- **Elastic fibers:** Provide recoil that resists stretch and drives passive expiration.

- **Surfactant:** Secreted by type II alveolar cells, it reduces alveolar surface tension, preventing collapse by reducing surface tension, and increasing compliance.
- **Age** affects the curve, with infants having more compliant chest walls but less compliant lungs compared to adults.



CLINICAL CORRELATION

Surface tension is the attractive force between liquid molecules at the air-liquid interface that minimizes surface area, and according to the Law of Laplace ($P = 2T/r$), the **smaller the alveolar radius (r)**, the **greater the pressure (P)** required to keep it open for a given surface tension (T). Here is a 5-second video demonstrating this LaPlace effect, where a smaller balloon empties into a larger one (see video below).

Significantly, surfactant reduces surface tension on the surface of the alveoli, thus preventing **small alveoli from collapsing into larger ones**.



Chest wall compliance

Unlike the lungs, which recoil inward, the chest wall has its own neutral equilibrium point, around 55–60% of total lung capacity ($\approx 75\text{--}80\%$ of vital capacity), where it neither expands nor recoils (**Figure 7**). Below this volume, the chest wall tends to spring outward, and above it, it tends to recoil inward. On the pressure–volume curve, this is reflected as negative pressures (outward pull) below the equilibrium point (0) and positive pressures (inward push) above it (**Figure 7**). This means that at low lung volumes (below FRC), the chest wall exerts an outward force, while at high lung volumes, it exerts an inward force.

Chest wall compliance is affected by various factors. It decreases with **aging** due to stiffening of the ribs and intercostal muscles, and it is further reduced by obesity, kyphoscoliosis, neuromuscular weakness, pleural diseases, or trauma that limits chest expansion. In contrast, infants have a

highly compliant chest wall, offering little resistance to the lungs' inward recoil, making them more prone to alveolar collapse and respiratory fatigue. Changes in chest wall compliance shift the pressure–volume curve and increase the **work of breathing**, influencing overall respiratory mechanics.

Respiratory System Compliance (Chest and Lung combined)– and Functional Residual Capacity (FRC) (LO2)

The **balance between the outward recoil of the chest wall and the inward elastic recoil of the lungs** affects the total respiratory system compliance. As indicated above the lungs want to recoil outwards, or to 70% of lung volumes when pressures are 0. The lungs, on the other hand, collapse to the residual volume when the inflation pressures are near zero.

The **Functional Residual Capacity (FRC)** is the lung volume at which the respiratory system is at mechanical equilibrium between the lungs and chest wall (Figure 7). At FRC, the inward recoil or "pull" of the lungs is exactly matched by the outward expansion tendency of the chest wall, resulting in zero net pressure when alveolar pressure equals atmospheric pressure (with the glottis open). This stable resting volume (no muscular effort is needed) provides the foundation for normal tidal breathing. You will observe that there is no effort needed. At FRC, the respiratory compliance is most favorable, requiring the least work of breathing. When starting inhalation at higher FRC, the compliance worsens, and increases work of breathing.

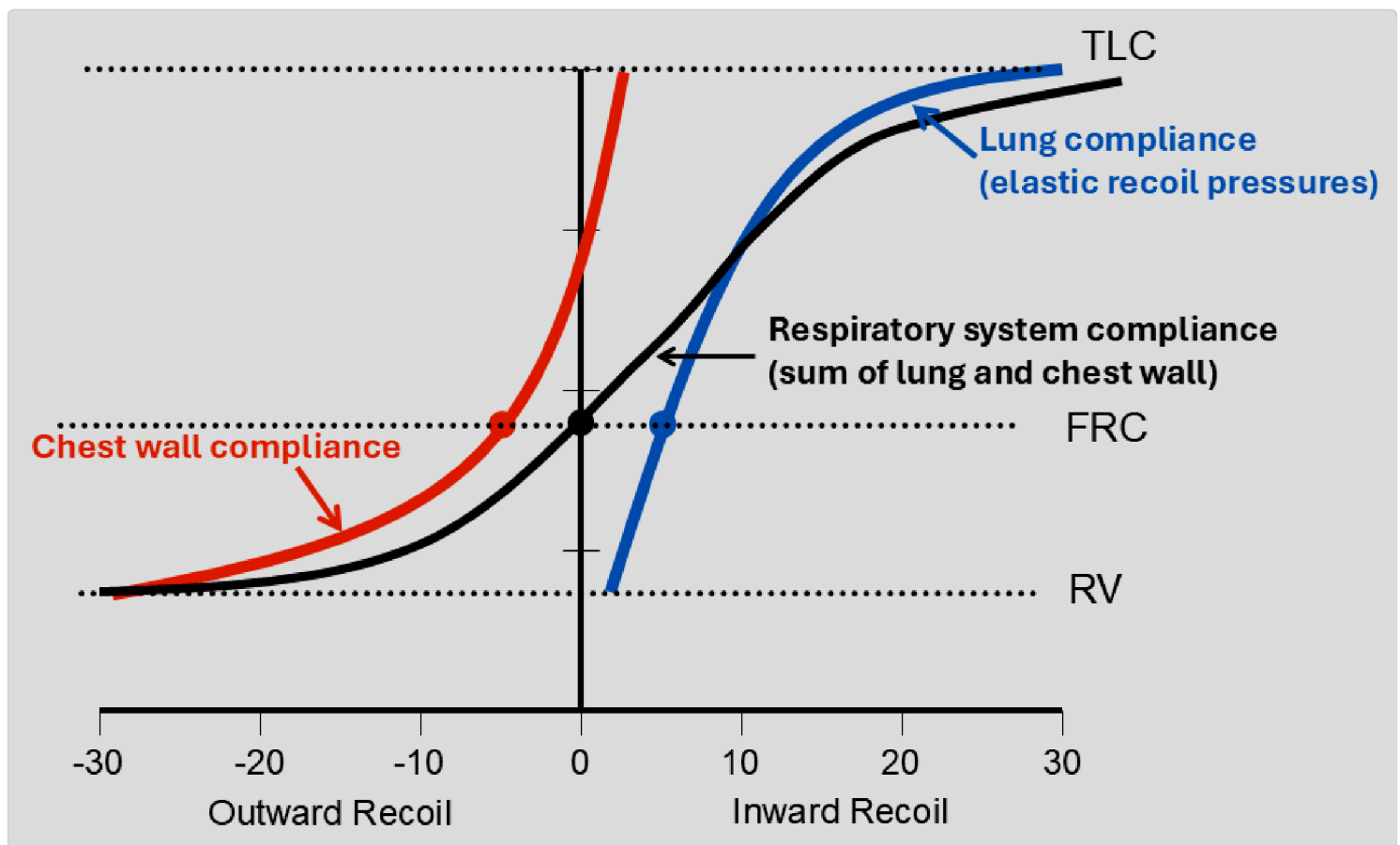


Figure 7. Lung and Chest wall compliance.

pressure (0 on the x-axis)

At lung\chest wall curve intersection with alveoli at atmospheric

Functional Residual Capacity– changes with lung compliance

Changes in the FRC, typically due to changes in lung compliance, can significantly impact respiratory function. In **restrictive diseases**, the FRC is typically reduced due to decreased lung compliance (the lung compliance curve shifts down and to the right). The equal and opposing chest wall force, also occurs lower down on that curve, decreasing the FRC (**Figure 8**, black dot is lower). In **obstructive diseases** like emphysema, lung compliance curve shifts upwards due to loss of elastic recoil. The curve moves upward and to the left, increasing compliance and raising FRC (**Figure 8**, black dot is higher). The FRC is often increased due to air trapping and hyperinflation. Identifying these changes through analysis of pressure-volume curves provides valuable diagnostic information and guides therapeutic interventions.



CLINICAL CORRELATION

Why is it beneficial to keep a volume of air in our lungs at end exhalation (FRC), instead of exhaling all of our lung volume with each breath?

Functional residual capacity (FRC) represents the ideal operating point of the lungs, where the opposing elastic forces of the lungs (inward) and chest wall (outward) are balanced. With an open glottis, alveolar pressure equals atmospheric pressure at FRC.

FRC serves several important physiologic roles. It acts as a large reservoir of oxygen—containing enough to sustain gas exchange for multiple breaths—which helps maintain oxygenation between inspirations. By keeping alveoli partially inflated, FRC prevents alveolar collapse, thereby reducing shunt and hypoxemia. At this lung volume, pulmonary vascular resistance is near its minimum, resulting in relatively high pulmonary blood flow and favorable ventilation–perfusion matching. In addition, lung compliance is greatest around FRC, making it the most efficient starting point for tidal breathing and minimizing the work of breathing.

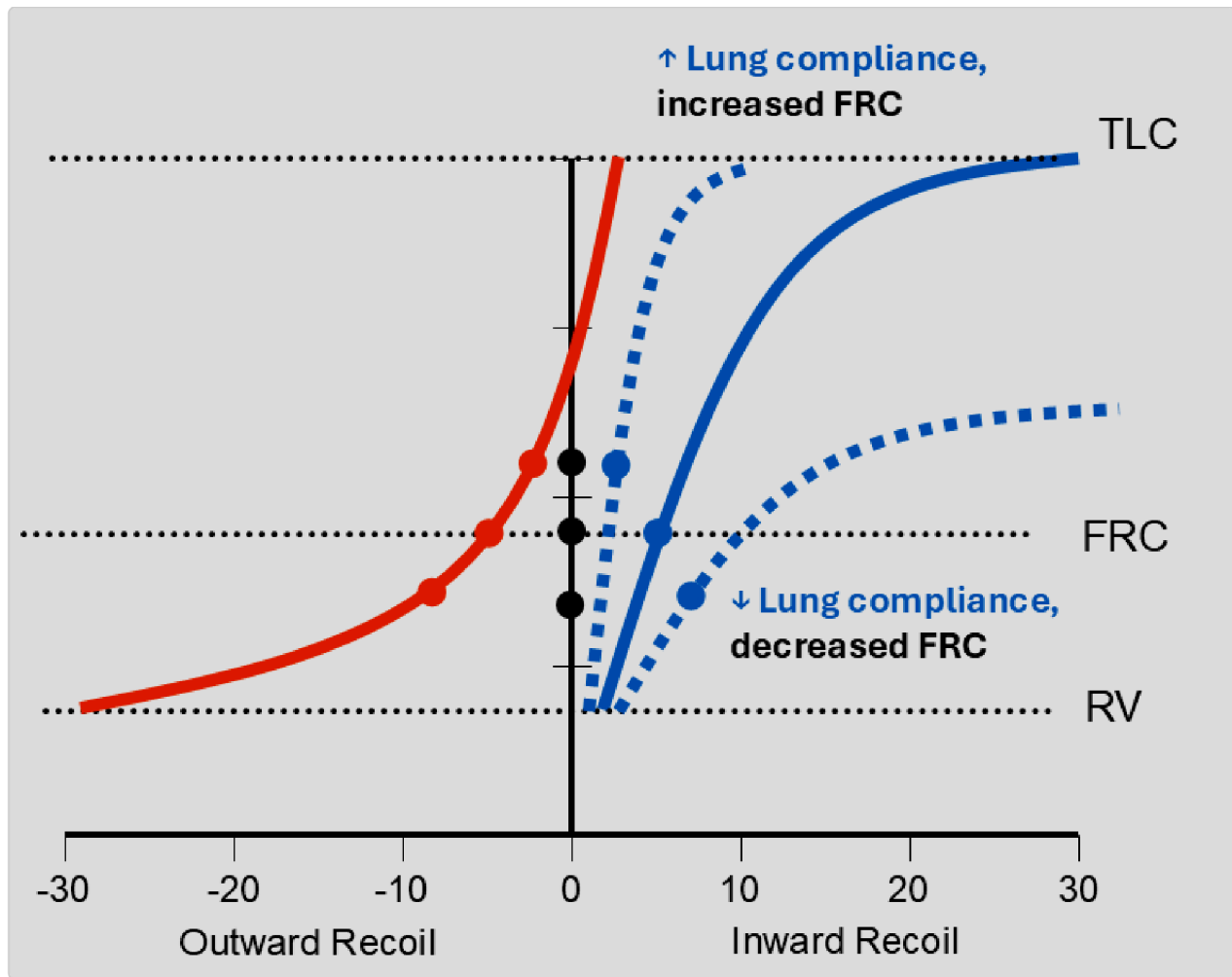


Figure 8. How changes in Lung compliance alters FRC in the combined system

At chronically higher lung volumes, the respiratory muscles have to work harder to overcome the less compliant, over-inflated lungs. This contributes to the feeling of breathlessness and discomfort. The FRC is where the entire respiratory compliance is most favorable. Higher FRC means worse compliance and increased work of breathing.



CASE CONNECTION

[BACK TO INTRODUCTION](#) ↑

JW's symptoms, exam findings, and pulmonary function tests are characteristic of obstructive lung disease, specifically COPD with emphysema. The decreased

FEV₁ and FEV₁/FVC ratio, along with hyperinflation and elevated residual volume, reflect air trapping and increased airway resistance, especially during expiration. Loss of alveolar elastic recoil weakens radial traction on the small airways, making them more prone to dynamic airway collapse during forced expiration. This contributes to expiratory airflow limitation, or dynamic airway collapse, which contributes to the characteristic “scooped-out” appearance on the PFT flow-volume loop. This also explains JW’s prolonged exhalation and use of accessory muscles when breathing.

His increased lung compliance results in a leftward/upward shift in the pressure-volume curve, but the trade-off is a loss of recoil, which reduces the driving force for expiration. The resulting air trapping elevates his functional residual capacity (FRC) and causes a barrel-chested appearance on imaging. His expiratory wheezes arise from turbulent flow through narrowed and collapsible airways, while his hypoxemia reflects ventilation-perfusion mismatch. Together, these physiologic derangements explain JW’s progressive dyspnea and limited exercise tolerance.

Summary

- Airway resistance is most dramatically affected by airway radius, with resistance inversely proportional to the fourth power of radius (Poiseuille's law), explaining why even small airway narrowing causes significant clinical effects.
- Airflow patterns influence resistance, with turbulent flow requiring more energy and creating higher resistance than laminar flow, especially in larger airways, at branching points, and during forced breathing maneuvers.
- Lung volume directly affects airway resistance through radial traction—as lung volume increases during inspiration, airways are pulled open,

decreasing resistance; conversely, resistance increases during expiration as lung volume decreases.

- Radial traction forces depend on the integrity of lung connective tissue and degree of lung inflation; disruption of these forces in diseases like emphysema contributes to dynamic airway collapse and increased resistance.
- Lung compliance ($\Delta V/\Delta P$) is due to elastic recoil and surfactant; steeper slopes indicate greater compliance (more distensible lungs). Lung compliance
- Chest compliance changes with lung volume; with outward expanding forces during normal breathing volumes.
- The functional residual capacity (FRC) occurs at the point where opposing elastic recoil forces of the lungs (inward) and chest wall (outward) are in equilibrium, preventing alveolar collapse between breaths.
- Pathological conditions alter pressure-volume relationships: restrictive diseases shift the curve downward/rightward (decreased compliance), while obstructive diseases like emphysema shift it upward/leftward (increased compliance but reduced elastic recoil).

Review Questions

1. What effect does turbulent airflow have on airway resistance compared to laminar flow?

- A. Decreases resistance
- B. No change in resistance

- C. Increases resistance markedly
- D. Occurs only in small airways
- E. Occurs only during quiet breathing

Explanation (requires correct answer)

2. During which phase is airway resistance greatest in emphysema with increased lung compliance?

- A. Normal inspiration
- B. Forced expiration
- C. Normal expiration
- D. Breath-holding
- E. Forced inspiration

Explanation (requires correct answer)

3. What is radial traction's main role in airway resistance?

- A. Narrows airway radius
- B. Causes smooth muscle contraction
- C. Increases mucus production
- D. Pulls airways open
- E. Reduces lung volume

Explanation (requires correct answer)

A.

B. Lung curve inflection point

- C. Lung/chest wall curve intersection
- D. Zero transpulmonary pressure
- E. Maximum chest wall compliance

Explanation (requires correct answer)

5. During normal, quiet breathing, which of the following structures will most likely demonstrate turbulent flow?

- A. Alveolar ducts
- B. Bronchi
- C. Bronchioles
- D. Terminal bronchioles
- E. Trachea

Explanation (requires correct answer)

How would you rate this Brick?



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01

Explain how airway resistance is affected by airway radius, turbulent flow, lung compliance, and lung volume (including La Place)

02

Explain the pressure-volume (compliance) curves for the lungs, chest wall, and respiratory system (including the FRC)

Objective Confidence: 0/2